

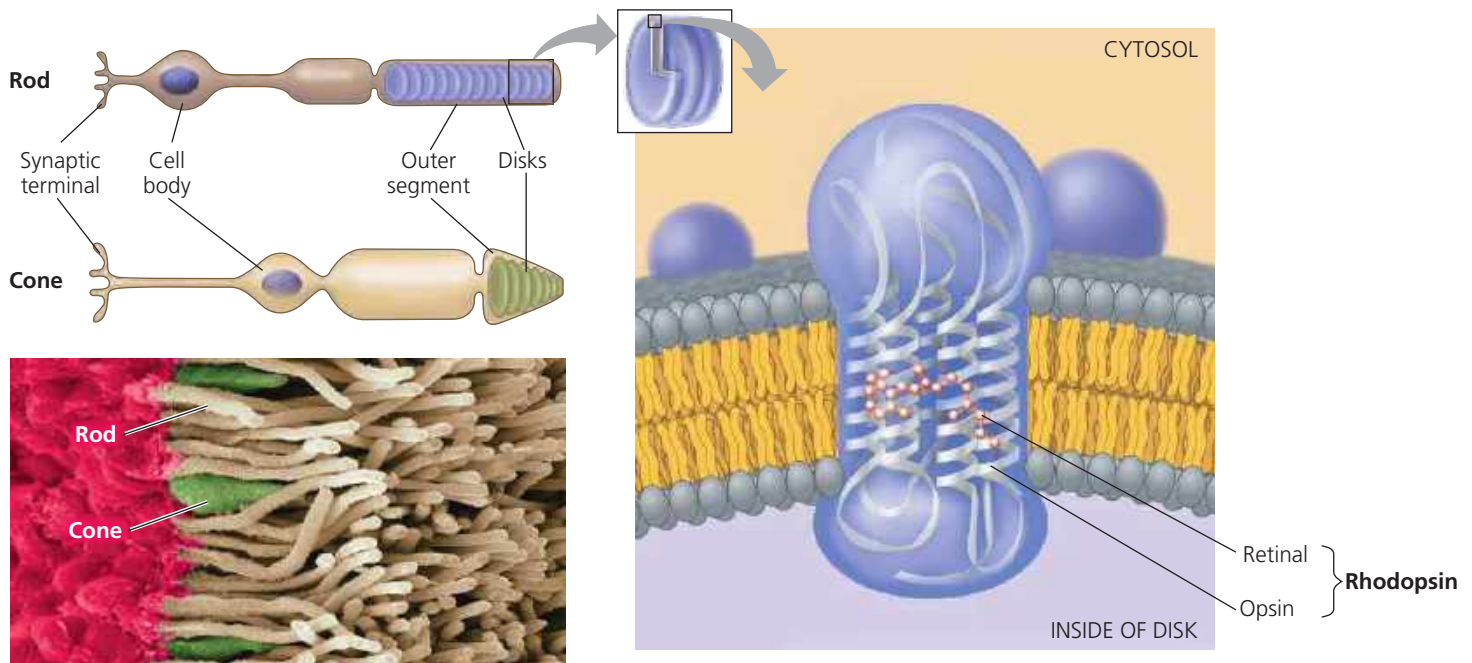
The eyes of all vertebrates have a single lens. In fishes, focusing occurs as in invertebrates, with the lens moving forward or backward. In other species, including mammals, focusing is achieved by changing the shape of the lens.

The Vertebrate Visual System

The human eye will serve as our model of vision in vertebrates. As described in **Figure 50.17**, vision begins when

photons of light enter the eye and strike the rods and cones. There the energy of each photon is captured in retinal, the light-absorbing molecule in the visual pigment rhodopsin.

Although light detection in the eye is the first stage in vision, remember that it is actually the brain that “sees.” Thus, to understand vision, we must examine how the capture of light by retinal changes the production of action potentials and then follow these signals to the visual centers of the brain, where images are perceived.



3 Photoreceptor Cells

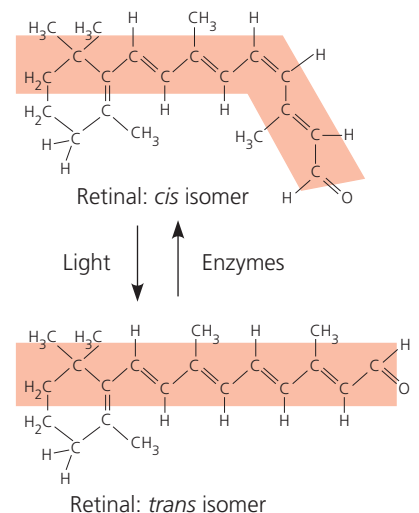
Humans have two main types of photoreceptor cells: rods and cones. Within the outer segment of a rod or cone is a stack of membranous disks in which *visual pigments* are embedded. **Rods** are more sensitive to light than cones are, but they do not distinguish colors; rods enable us to see at night, but only in black and white. **Cones** provide color vision, but, being less sensitive, contribute very little to night vision. There are three types of cones. Each has a different sensitivity across the visible spectrum, providing an optimal response to red, green, or blue light.

In the colorized SEM shown above, cones (green), rods (light tan), and adjacent neurons (red) are visible. The pigmented epithelium, which was removed in this preparation, would be to the right.

4 Visual Pigments

Vertebrate visual pigments consist of a light-absorbing molecule called **retinal** (a derivative of vitamin A) bound to a membrane protein called an **opsin**. Seven α helices of each opsin molecule span the disk membrane. The visual pigment of rods, shown here, is called **rhodopsin**.

Retinal exists as two isomers. Absorption of light shifts one bond in retinal from a *cis* to a *trans* arrangement, converting the molecule from an angled shape to a straight shape. This change in configuration activates the opsin protein to which retinal is bound.



➔ **Mastering Biology Animations: Structure and Function of the Eye • Function of the Retina • Photoreception**

VISUAL SKILLS The isomers of retinal have the same number of atoms and bonds but differ in the spatial arrangement at one carbon-carbon double bond ($C=C$). In each isomer, circle that ($C=C$). Looking at the atoms around that bond, to what atoms do the terms *cis* (same side) and *trans* (opposite side) refer?